

ECON100B - SECTION 21

Wrap up of IS-LM-PC and Intro to Stabilization Policy

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- PSET 3 due on Friday, 4/17
- PSET 4 released next week

Agenda

- Stabilization Policy
- Intro to Government

Stabilization Policy



The challenge of policymaking

- In the U.S., the federal reserve has a dual mandate:
- Keep inflation down (PC model)
- Keep employment high (IS model)
- This is an extremely difficult (impossible?) dual mandate. Why?
- The central bank does their best to meet this dual mandate through influencing the real interest rate (MP model)

Monetary Policy Rules

- Monetary policy rules are set of instructions that determine the stance of monetary policy during a given situation
- For example: we should raise interest rates when inflation is high OR we should lower interest rates when short run output is low

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi}).$$

- What is this rule saying?

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$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi}).$$

- What is this rule saying?
- Raise interest rates when inflation is high, lower interest rates when inflation is low
- Target inflation rate of 2% across many of the largest global economies

Why Pursue Monetary Policy Rules

- Monetary policy rules are important because they help anchor decision making by central banks
- Put trust into the system, help to make the economy more stable over time
- Help individuals, businesses, and investors know what to expect for economic outcomes
- In reality: the U.S. Federal Reserve doesn't officially have "Monetary Policy Rules", but it is rather implied
- Balance of Rules and Discretion

Monetary Policy under Financial Crises (Ch. 14)

- Suppose we enter a period of economic turmoil, where consumers reduce their spending
- How would we expect the federal reserve to respond to this shock?

Monetary Policy under Financial Crises (Ch. 14)

- Suppose we enter a period of economic turmoil, where consumers reduce their spending
- How would we expect the federal reserve to respond to this shock?
- We would expect the federal reserve to drop interest rates to encourage more investment
- But consider the fact that the real interest rate observed in the economy includes a “risk premium”
- What happens if the “risk premium” increases as banks become more skeptical about lending?

Putting it all together: IS-MP-PC Model

3. (36 points total) *Thanks a lot, Generation X, You Broke the Short Run.*

“Generation X” is the term for Americans born roughly between the 1960s and 1980s. People born later are commonly called “Millennials.” Recall the standard IS, MP, and PC Curves we saw in class:

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \quad (4)$$

$$R = R^{ff} + \bar{p} \quad (5)$$

$$\Delta\pi = \bar{v}\tilde{Y} + \bar{o} \quad (6)$$

where R is the real interest rate paid by borrowers; $\bar{r}$ is the marginal product of capital; R^{ff} is the real federal funds rate; $\bar{p}$ is the risk premium charged by private lenders; π is the inflation rate; and $\bar{o}$ is an external inflation shock.

Imagine that it is 2002, and the economy has emerged from the recession of 2001. Generation X'ers feel optimistic about the future, and they increase spending on housing as a share of permanent income, racking up debt.

- (a) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.*

Putting it all together: IS-MP-PC Model

Now imagine that it is 2006, and a major crisis hits the housing market. Generation X'ers discover that their incomes are not sufficient to make their mortgage payments on the enormous houses they starting buying in 2002.

Homeowners sharply reduce their consumption spending in hopes of avoiding default on their mortgages. But it's too little, too late.

- (b) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from part (a).]

Putting it all together: IS-MP-PC Model

Despite their efforts, many Generation X'ers default on their mortgages, and lenders lose their loaned funds. Unfortunately, those “lenders” aren't just a few banks. Because the original mortgage lenders sold those mortgage agreements off to many different banks and investors, many agents in the economy lose assets. Even worse, remaining lenders don't know exactly which agents lost a lot in the mortgage market, only that many of them did and thus are not good borrowers because they're near bankruptcy.

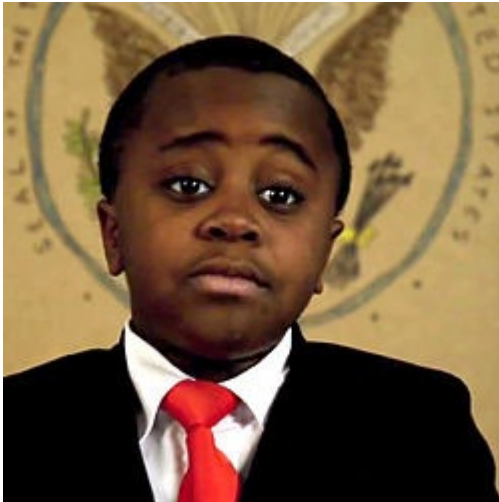
(c) (6 points) If you were a good lender during this period, with funds available, and a borrower walked into your office and asked to borrow money, what interest rate would you charge him or her? Would it be higher, lower, or the same as the usual interest rate you'd charge, and why?

(d) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from parts (a) or (b).]

Putting it all together: IS-MP-PC Model

- (e) (6 points) If the Federal Reserve had already reduced the nominal federal funds rate to zero before the events of part (d), would that change your story at all, and if so how? Explain.
- (f) (6 points) What else could the government use to address the issues afflicting the economy in part (d) other than the Federal Reserve's monetary policy? Illustrate this alternative in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. [You *should* show the events from part (d) but not from parts (a) or (b).]

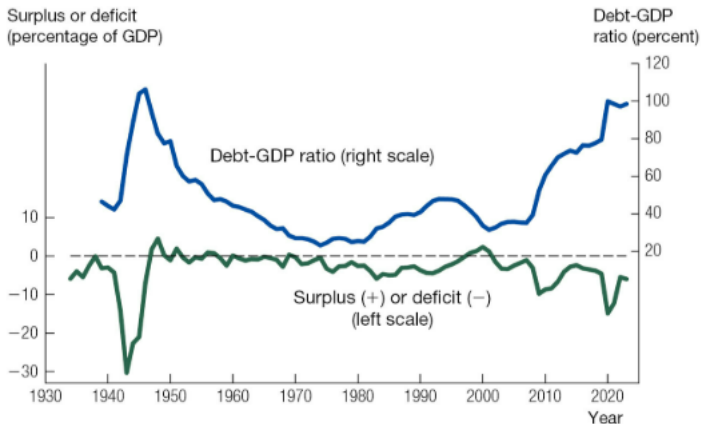
Government (Ch. 18)



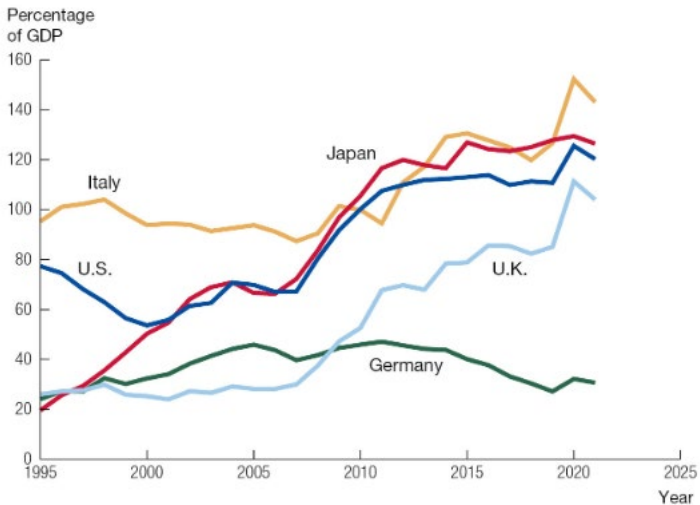
The government in the macroeconomy

- Federal spending typically hovers between 20-25% of the GDP so a huge component of the total macroeconomy
- Biggest categories of spending include healthcare, national defense, social security, and income security, and interest payments on debt (!)
- Because government spending in recent history has almost always been higher than tax revenues, the government uses debt financing to make up for the gap
- We can use the debt-GDP ratio to calculate the amount of debt owed vs the GDP of a country
- In the U.S., this has risen from 35% to 100% in the past 20 years
- (Almost) anyone can buy U.S. debt. Individual investors, institutional investors, foreigners, etc.

Debt GDP ratio



Debt-GDP Ratios around the World



The Government Budget Constraint

- The government must have a “balanced” budget, leading us to the following budget constraint:

$$\underbrace{G_t + Tr_t + iB_t}_{\text{uses}} = \underbrace{T_t + \Delta B_{t+1} + \Delta M_{t+1}}_{\text{sources}}.$$

- Uses: Government spending, transfer payments, and interest payments
- Sources: Tax revenue, new borrowing bonds, and new money changing the money supply (assume 0 given fiscal-monetary independence)

Simplifying the Budget Constraint

$$B_{t+1} = (1 + i)B_t + \underbrace{G_t - T_t}_{\text{deficit}}$$

- The stock of debt next year equals:
- The stock of debt this year, including interest payments
- The change in the government deficit based on spending minus transfer payments

Government Practice

1. What is the debt to GDP ratio in the United States? Has it been growing over time and how does it compare to other parts of the world?
2. Suppose I have \$5T in government spending and \$4.5T in tax revenue in a given year. Also suppose that I have a current debt of \$4T with a 5% interest rate. What will my debt be next year?

Government Practice Solutions

1. What is the debt to GDP ratio in the United States? Has it been growing over time and how does it compare to other parts of the world?
2. \$4.7T of debt. Solve via $\$4T * (1.05) + \$0.5T$